

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2016-17

Paper No. CHB-101: Structure & Bonding and Organic Chemistry-I

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Structure and Bonding – Marks: 35)

Question 1. Answer all parts of the following: [5 × 2 = 10]

- (a) Depict the radial distribution curve for a 4d orbital and mention the number of radial nodes in it.
- (b) Why are Group 1 metals quite soft as compared to transition metals? Explain.
- (c) What is the radius ratio rule? In what way does it help in predicting the coordination number of ionic crystals?
- (d) Arrange the following in order of increasing covalent character, giving appropriate reason: NaCl, MgCl₂, AlCl₃.
- (e) Calculate the formal charge of each atom in the nitrite ion (NO₂⁻).

Question 2.

- (a) State Schrödinger wave equation and discuss the physical significance of ψ and ψ^2 . [6]
- (b) Draw the molecular orbital energy level diagram for NO and mention the HOMO and LUMO. Calculate its bond order. [6]

Question 3.

- (a) With the help of suitable examples, discuss van der Waals interactions. Are these stronger or weaker than hydrogen bonding? [6]
- (b) Discuss Slater's rules for calculating the effective nuclear charge (Z_{eff}) and calculate Z_{eff} for a 3d electron in a zinc atom ($Z = 30$). [6]

Question 4.

- (a) Explain why beryllium exists in the metallic form and not as a diatomic gas molecule (Be_2) on the basis of Molecular Orbital Theory. [6]
- (b) Using VSEPR theory, predict the geometries and draw the structures of the following:
(i) SF_4 , (ii) ClF_3 , and (iii) XeF_2 . [6]

SECTION B

(Organic Chemistry-I – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

- (a) Why is the enol form of ethyl acetoacetate more stable than that of simple acetone?
- (b) Explain why methyl radical ($\cdot\text{CH}_3$) has a planar geometry whereas trifluoromethyl radical ($\cdot\text{CF}_3$) has a pyramidal geometry.
- (c) Draw the chair and boat conformations of cyclohexane and explain their relative stability.
- (d) What are enantiomers and diastereomers? Explain with one suitable example of each.
- (e) Briefly discuss the mechanism of $\text{S}_{\text{N}}2$ reaction with a suitable example.

Question 6.

- (a) How will you prepare the following compounds starting from ethyl acetoacetate? Write stepwise mechanisms: [3 × 4 = 12]

- (i) Ethyl methyl ketone
- (ii) *n*-Valeric acid
- (iii) Succinic acid
- (iv) Cyclopropanecarboxylic acid

OR

- (a) How are the following compounds prepared from Grignard reagent (RMgBr)? Write reactions with suitable mechanisms: [3 × 4 = 12]

- (i) Primary alcohol
- (ii) Carboxylic acid
- (iii) Ketone
- (iv) Alkane

Question 7. Explain any four of the following: [4 × 3 = 12]

- (a) 1,3-Butadiene gives 1,2- as well as 1,4-addition products with bromine.
- (b) Dehydration of alcohols generally needs an acid catalyst.
- (c) *cis*-Hydroxylation of 2-butene with cold alkaline KMnO_4 gives *meso*-2,3-butanediol.
- (d) Chlorination of methane in the presence of sunlight is a free-radical chain reaction.

- (e) S_N1 reactions are accompanied by racemization at the chiral center.

Question 8.

- (a) Draw structural formulae for each of the following pairs of compounds: [4]
- (i) (*E*)- and (*Z*)-2-pentene
 - (ii) (*E*)- and (*Z*)-2-phenyl-2-butene
- (b) How will you carry out the following conversions? Write mechanisms in each case: [4 × 2 = 8]
- (i) Cyclohexene to *trans*-1,2-cyclohexanediol
 - (ii) Propene to 1,2-dibromopropane